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High-resolution studies of rainfall on Norfolk Island Part II: Interpolation of rainfall data

K.N. Dirks[a,*], J.E. Hay[b], C.D. Stow[a], D. Harris[a]

[a]Department of Physics, The University of Auckland, Private Bag 92019, Auckland, New Zealand
[b]School of Environmental and Marine Sciences, The University of Auckland, Private Bag 92019, Auckland, New Zealand

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Abstract

Four spatial interpolation methods are compared using rainfall data from a network of thirteen rain gauges on Norfolk Island (area 35 km^2). The purpose is to obtain spatially continuous rainfall estimates across the island, from point measurements and for different integration times, by the most effective means. The more computationally demanding method of kriging provided no significant improvement over any of the much simpler inverse-distance, Thiessen, or areal-mean methods. In order to assimilate some of the characteristics of spatially varying rainfall, and based on the comparisons performed, the inverse-distance method is recommended for interpolations using spatially dense networks. © 1998 Elsevier Science B.V. All rights reserved.

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1. Introduction

One of the problems facing meteorologists and hydrologists studying spatial rainfall patterns is the interpolation of data from irregularly spaced rain gauges in order to determine mean areal rainfalls or to characterise rainfall variability within a region or catchment. Similar problems arise in the numerical interpolation of other meteorological parameters such as solar irradiances (Hay, 1986) or maximum and minimum temperatures (Mitchell, 1991). Rainfall is intermittent and spatially discontinuous, frequently with zero accumulations; for this reason the reliable spatial interpolation of rainfall is inherently more difficult than for many other variables. Over short integration times, interpolation may be impossible. The purpose of this paper is to determine the most efficient method of interpolation for the Norflok Island high-density rain gauge network. The method can then be used in studies of the effect of rainfall variability on island water resources and floods.

Many schemes with varying degrees of complexity have been proposed for the spatial interpolation of rainfall (Thiessen, 1911; Shepard, 1968; Delfiner and Delhomme, 1973). Considerable effort has gone into the quantitative evaluation of each of these schemes and into determining the optimum parameters required for some of the interpolation methods (e.g. Tabios and Salas, 1985; Boussières and Hogg, 1989). Studies have considered as many as hundreds of gauges over areas up to 10^5 km^2, and have used daily or monthly totals. The purpose of this paper is to evaluate four simple interpolation methods and determine their optimum parameters over a grid spacing of

* Corresponding author. E-mail: k.dirks@auckland.ac.nz

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6″ longitude (at this latitude equal to 187 m by 187 m) and over temporal scales ranging between hourly and yearly. The investigations are important since they provide optimum parameters and estimations of accuracy for the interpolation algorithms which may be applicable to high-density networks elsewhere. Moreover, such high-resolution data are needed to better understand the orographic and hydrological processes characteristic of a small island or catchment.

The rainfall data used in the present study are for Norfolk Island, a subtropical island 35 km^2 in area located in the South Pacific about halfway between New Caledonia and the northern tip of New Zealand at 167°E and 29°S (Stow and Dirks, 1998).

Four interpolation methods are evaluated: Thiessen, inverse-distance, areal-mean and kriging. In Part III (Bradley et al., 1998) of the series, interpolated rain fields are compared with predictions of a simple analytical model of rainfall caused by topographically modified wind flow. Although almost all of the data described here were obtained using a sampling time of 15 s, the shortest time integration used in the present studies is one hour because typical transit times across the island for moving rain systems are of the order of 10 minutes. The restriction to one-hour-or-greater time averages for the analysis precluded the need for the consideration of time delays between gauge locations and points of interpolation.

2. Interpolation methods

Thiessen (1911) first considered the problem of how to infer areal rain over a grid from point measurements. A method was proposed which involved assigning a value to each grid location equal to the value of the closest gauge. The resulting interpolated field consists of polygons with the boundaries at a distance halfway between gauge pairs. Another commonly used technique, areal mean interpolation, involves assigning a grid-point value equal to the mean of all the gauge readings within the array. In this method, each gauge is weighted equally for every interpolated point. In both interpolation methods, information on rainfall gradients is lost. In order to overcome this problem, hydrologists have developed methods that incorporate the spatial variation of rainfall within a catchment into the interpolation scheme. Many of these schemes are based on some type of inverse-distance weighting function (Cressman, 1959; Shepard, 1968; Barnes, 1973) but each scheme differs in the form of the weighting function. The method of Barnes uses an exponential function of distance rather than a power function. The method of Cressman gives a significant weighting to distant gauges, while the method of Shepard assigns greater relative importance to the closest gauges. In all schemes, the sum of the weights for an interpolated point must be unity.

Although the inverse distance schemes show spatial trends in rainfall, they are limited in that the powers of the weighting functions need to be selected before the interpolation may be performed. This choice of power may significantly affect the accuracy of the resulting interpolated field. The method of kriging (Davis, 1986) avoids the need to preselect parameters. In this method, the semivariance is calculated for each pair of gauges in the network and plotted against gauge separation distance. The semivariance becomes a measure of the degree of spatial dependence between points along a transect and is used to determine the weighting function. The method of optimal interpolation is similar to kriging, except that the weighting function is determined from the correlation between gauge readings, in the form of a correlogram, rather than from a semivariogram. The method of optimal interpolation is not evaluated here, since the results have been shown to be operationally indistinguishable from kriging (Tabios and Salas, 1985; Creutin and Obled, 1982).

There have been many comparative assessments of common interpolation schemes. Tabios and Salas (1985) found that for monthly totals for a large-scale network, the statistical methods of kriging and optimal interpolation performed better than the Thiessen and inverse distance schemes. Boussières and Hogg (1989) found similar results in a study of daily totals comparing optimal interpolation to inverse distance schemes. Creutin and Obled (1982) found that, for low- to medium-density networks, kriging performed better than the Thiessen or areal mean methods.

It appears from the foregoing studies that, for low-density networks, the statistical methods produce superior results to those of simpler interpolation techniques. However, most of the comparative studies of interpolation schemes have used spatial scales of the order of thousands of square kilometres. The importance of the present study is that it is being conducted on a spatial scale which is at least an order of magnitude smaller. This evaluation of the common schemes has been undertaken using Norfolk Island rainfall data in order to investigate the comparative performance of rudimentary and more sophisticated interpolation procedures for a high-resolution rain gauge network. Thus, in spite of the former being largely obsolete, such methods are evaluated in order to provide a basis for quantitative comparison. An outline of the method used for assessing each of these schemes quantitatively is given below.

3. Evaluation of interpolation schemes

The effectiveness of each of the interpolation schemes (Thiessen, areal-mean, inverse-distance, kriging) was evaluated using the fictitious-point method (or cross-validation) first proposed by Seaman (1983) and subsequently used by Delhomme (1978) and Boussières and Hogg (1989). The error of interpolation in the fictitious-point method can be defined by the root-mean-squared error, σ_{f}, given by

$$\sigma_{\mathrm{f}} = \sqrt{\frac{\sum_{j=1}^{n} (r_j - \hat{r}_j)^2}{n}}, \tag{1}$$

where r_j is the observed rainfall and $\hat{r}_j$ is the interpolated rainfall estimate (using $n-1$ values, excluding j) at the jth gauge location. The coefficient of variation, ρ_{f}, is then defined by:

$$\rho_{\mathrm{f}} = \frac{\sigma_{\mathrm{f}}}{\frac{1}{n}\sum_{j=1}^{n} r_j} \tag{2}$$

ρ_{f} is used to compare the performance of interpolation schemes for different integration times.

Data from each of the years 1991, 1992 and 1993 were treated separately due to the differences in the number of operating gauges from year to year (eight, 11 and 13 gauges, respectively). For each of the integration times evaluated, three yearly-average ρ_{f} values were determined. In Fig. 1, the average ρ_{f} values are plotted as a function of integration time, the error bars corresponding to the standard deviations. Note that there are no error bars on the yearly-total integrations since the values of ρ_{f} are constructed from a single interpolated field. In the figure, the points corresponding to the three sets of data have been offset horizontally so that the corresponding error bars can observed more easily. The 1991 data have been shifted slightly to the left, while the 1993 data have been shifted slightly to the right relative to data points for the year 1992.

Fig. 1a presents the average ρ_{f} as a function of integration time for the Thiessen interpolation method. The errors ρ_{f} decrease substantially with increasing integration time, values ranging from apprixmately 0.7 for hourly totals to about 0.2 for yearly totals. Results from this scheme are promising, considering its simplicity. Fig. 1b shows the interpolation errors for the areal-mean scheme. The results from this method are almost identical to those produced by the Thiessen method.

The inverse-distance interpolation scheme is more complex than the previous two schemes, since the power of the inverse distance function must be chosen before the interpolation is performed. A low power (<2) results in a greater contribution towards a grid point value of rainfall from distant gauges. As the power tends to zero, the interpolated field will approximate the areal-mean method, while for large values the scheme approximates the Thiessen method. An objective in the present study was to optimise this power of the inverse-distance function for various integration times, using Norfolk Island data for use in further work (see, for example, Part III). This was achieved by using the fictitious-point method to calculate ρ_{f} for a range of powers, p, for each of the integration times of interest.

The optimum power was obtained from the minimum on a plot of the average ρ_{f} versus the power. Fig. 2 provides an example of this plot for each of the 3 years, using a monthly-total integration. This figure shows that the exact choice of the numerical value of the power has minimal effect on the resulting errors providing the value is within the range of about 1.5 to 4. The conventional choice of $p = 2$ would be suitable for monthly totals. The value of ρ_{f} is higher, on average, for the year 1991 than for 1992 and 1993. However, this is not consistently the case for other integration times.

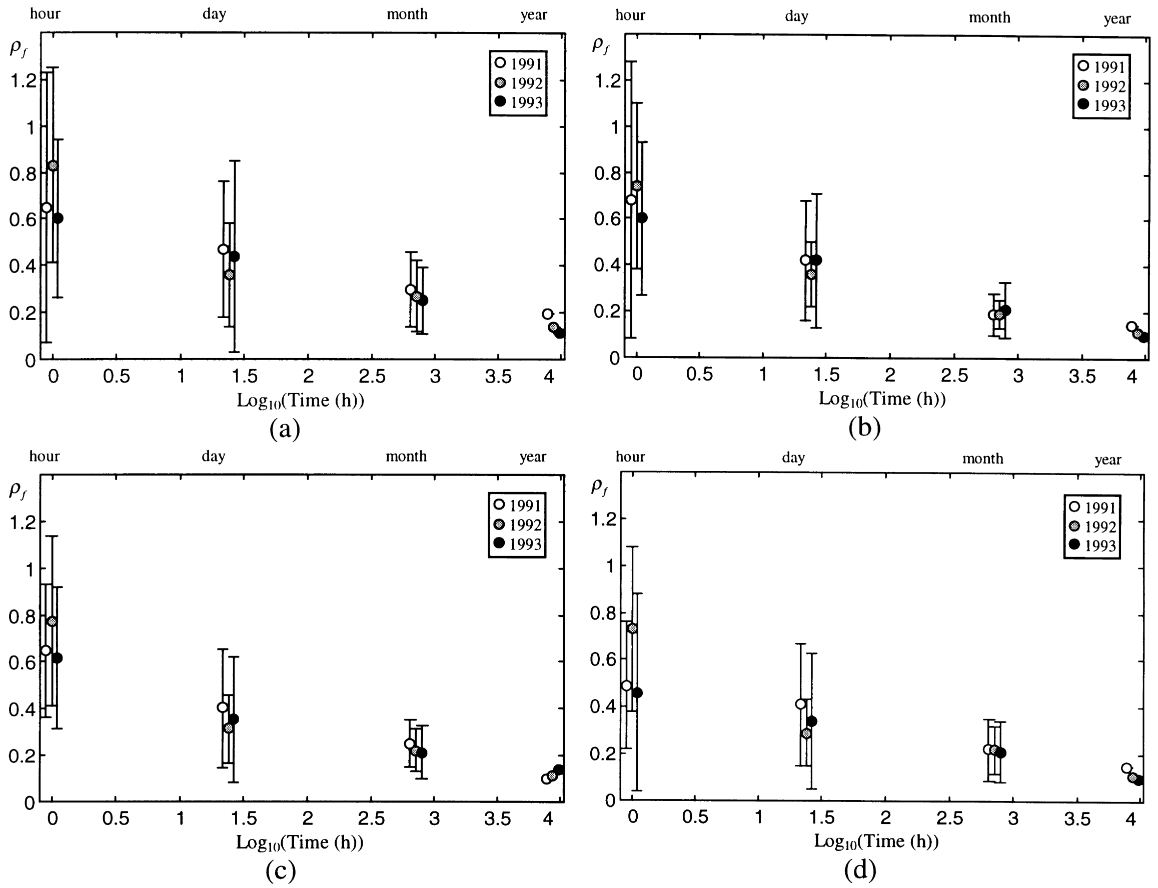

Fig. 1. The average ρ_f as a function of the log of the integration time for (a) Thiessen, (b) areal-mean, (c) inverse distance and (d) kriging.

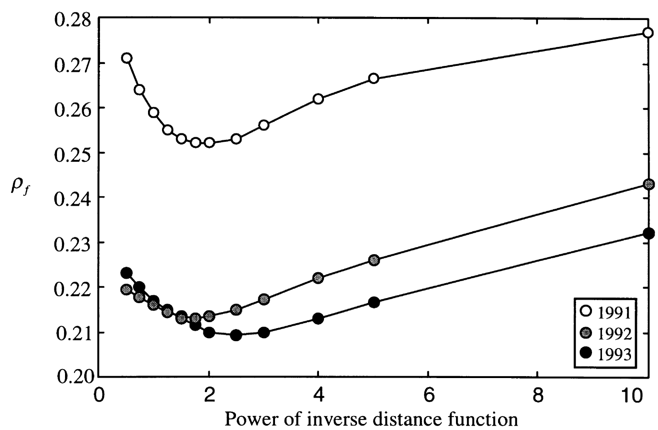

Fig. 2. ρ_f versus the power of the inverse-distance function for the monthly totals of 1991, 1992 and 1993.

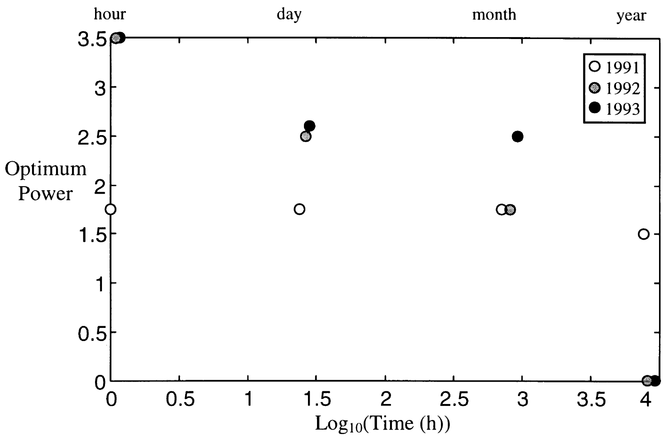

Fig. 3. The optimum power for the inverse-distance function as a function of the log of the integration time.

The optimum powers for the inverse distance scheme were evaluated for hourly, daily and yearly totals, in addition to the monthly totals discussed above; these are shown in Fig. 3. This figure may be used to pre-select the power when interpolating rainfall for a particular integration time. It can be seen that the optimum power decreases substantially with increasing integration time for the years 1992 and 1993, but not for 1991. Diagrams similar to Fig. 2 for other integration times reveal that the choice of a specific power value has relatively little effect on the magnitude of the interpolation error. Thus a strategy which used a power value of 2 for daily and monthly integration, 3 for hourly and 1 for yearly would appear to minimise the interpolation errors although the penalty for using a power of 2 throughout would not be great.

A low power for an inverse distance weighting function indicates a low spatial variability, and vice versa. It is apparent from Fig. 3 that the rainfall events on an hourly and daily basis were less variable for 1991 than 1992 or 1993. The reverse is true for yearly totals. The reason for this inconsistency is unclear. Many more years of data would be required to determine the cause of the variation, but is probably due to the highly variable nature of rainfall on a year-to-year basis.

The value of ρ_f was determined for each integration time and year using the optimum power. The resulting values of ρ_f were found to be almost identical to those found using either the areal-mean or Thiessen methods, as shown in Fig. 1c. Although the interpolation errors for this method are comparable to other simpler methods, the inverse-distance scheme produces rain fields that appear more realistic, making the interpolated fields potentially more useful in studies of the spatial distribution of rainfall.

The first step in evaluating an interpolated field using kriging is to produce the semivariogram. Fig. 4 shows the semivariogram produced for daily totals for the 3 years of observation. Similar, very noisy semivariograms were produced for the other integration times. The success of the kriging method depends on the assumption of a perfect semivariogram. This assumption is clearly violated, both in terms of the interpolated field and the associated variance. For this reason, and for consistency with the evaluation of the other three interpolation methods, the evaluation of the kriging method was undertaken using ρ_f rather than the variance field produced by the method itself. Based on Fig. 4, there appears to be no reason to suggest that the distance between pairs of gauges is ever large enough to reach the span (the separation between gauges beyond which the readings become essentially uncorrelated). This is not particularly surprising considering the small spatial extent of the network, which was geographically constrained by the size of the island.

The appropriate choice of the value of the semivariance at a distance of zero corresponding to the variance associated with small-scale variability and measurement errors was also difficult to evaluate. Both contribute to a non-perfect correlation at zero

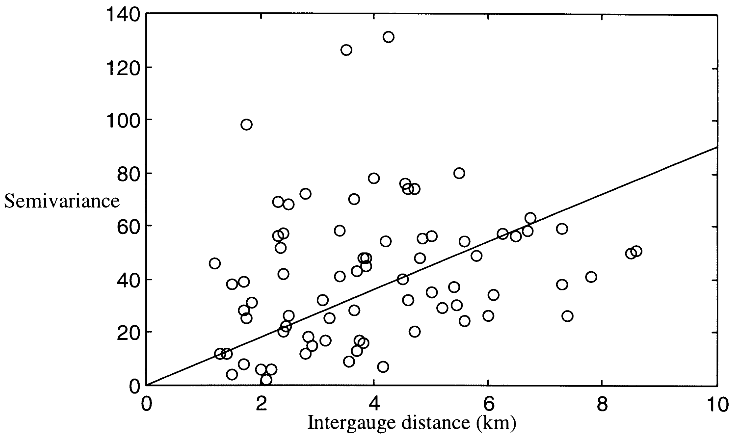

Fig. 4. The semivariogram for daily totals.

distance—the 'nugget effect' (Davis, 1986). In order to estimate the contribution of this error to the overall ρ_f statistic, an attempt was made to estimate the measurement errors and small-scale variability by comparing daily totals measured using one of the high-resolution rain gauges and a standard gauge placed approximately 6 m apart. Measurement errors and small-scale variability were found to be approximately one-third of the total error. These errors would impose a lower limit on the accuracy of a perfect interpolation scheme. Given the uncertainty of the 'nugget effect', the simple choice of a line of best fit passing through the origin was adopted for the semivariogram rather than the more commonly used nonlinear spherical fit. The results of assessing the accuracy of kriging are shown in Fig. 1d. As expected, ρ_f decreases with increasing integration time. Note that for hourly interpolations, even the method of kriging produces errors of about 0.6. On average, interpolation for yearly totals have errors of approximately 0.2.

4. Conclusions

The preceding data allow comparison of the errors related to the four interpolation schemes. Table 1 summarises the results for the four methods for the 3 years of observation. In the case of the inverse-distance interpolation, the optimum power for each integration time was used while for the method of kriging the semivariograms were used.

From Table 1, it can be concluded that all of the methods investigated perform in a comparable manner, within the range of variability observed. Given these results, the degree of complexity and the required amount of computer time, the kriging method does not appear to justify its use with this high-resolution network. The fact that the spatial extent of the network was such that the semivariograms did not reach the span for this particular network may well be the reason why the kriging method does not show significantly greater predictive skill than the simpler techniques. The Thiessen method obviously provides an unrealistic discontinuous rain field, and the areal-mean method clearly does not show the true spatial variation of rainfall. The inverse-distance method is considered to be the most appropriate choice for operational use for integration times of one hour or greater and the exact choice of the optimum power was not found to be crucial. It allows for variability studies, produces somewhat realistic rainfall fields and requires very little computational effort.

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Table 1
Errors of interpolation, average ρ_f, found for the four methods and four integration times investigated

		1991	1992	1993
Thiessen	Year	0.20	0.14	0.11
	Month	0.30 ± 0.16	0.27 ± 0.15	0.25 ± 0.14
	Day	0.47 ± 0.29	0.36 ± 0.22	0.44 ± 0.41
	Hour	0.65 ± 0.58	0.83 ± 0.42	0.60 ± 0.34
Areal-mean	Year	0.14	0.11	0.10
	Month	0.19 ± 0.09	0.19 ± 0.06	0.21 ± 0.12
	Day	0.42 ± 0.26	0.36 ± 0.14	0.42 ± 0.29
	Hour	0.68 ± 0.60	0.74 ± 0.36	0.60 ± 0.33
Inverse-distance	Year	0.10	0.11	0.13
	Month	0.25 ± 0.10	0.22 ± 0.09	0.21 ± 0.11
	Day	0.40 ± 0.25	0.31 ± 0.14	0.35 ± 0.26
	Hour	0.64 ± 0.28	0.77 ± 0.36	0.61 ± 0.30
Kriging	Year	0.15	0.11	0.10
	Month	0.22 ± 0.13	0.22 ± 0.10	0.21 ± 0.13
	Day	0.41 ± 0.26	0.29 ± 0.14	0.34 ± 0.29
	Hour	0.49 ± 0.27	0.73 ± 0.35	0.46 ± 0.42

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